

Sampling Plan: ADHD Prevalence in Aguascalientes

Homework 2: Sampling

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1. Problem Statement

This study looks at how common ADHD is among children and teenagers in the state of Aguascalientes, Mexico. The goal is to get a reliable estimate of ADHD screening rates in school-age students, which can help schools and health authorities plan better support and resources.

2. Population

All children and teenagers between 6 and 17 years of age enrolled in elementary and high schools in the state of Aguascalientes.

3. Variable to Measure

Whether the student screens positive for ADHD (yes or no). This is a categorical, binary variable.

4. Unit of Observation

Each individual student between 6 and 17 years old, enrolled in a school in Aguascalientes, on whom an ADHD screening result (positive or negative) is recorded.

5. Sampling Frame

The *Catálogo de Centros de Trabajo de la SEP*, which lists every public and private elementary and high school currently active in Aguascalientes. Once a school is selected, its own enrollment lists per group are used as the frame for the second stage.

6. Sampling Unit

Two sampling units are used, one for each stage of the design:

- **First stage:** the school. Each school registered in the SEP catalog is a sampling unit.
- **Second stage:** the group (classroom). Within each school selected in the first stage, groups are the sampling units from which students are drawn.

7. Sampling Method

Probabilistic sampling. Every school and every group has a known, nonzero chance of being selected, which allows the results to be generalized to the full population and lets us calculate a margin of error.

8. Type of Sampling

Two-stage stratified sampling.

- **Stage 1 (schools):** Schools are first grouped into strata, for example by municipality and by school type (public or private). A random sample of schools is then drawn from each stratum.
- **Stage 2 (groups):** Within each selected school, groups are randomly sampled, and all students in the chosen groups are screened for ADHD.

This design keeps costs and travel down while still giving fair representation to the different types of schools in the state.